Euclid's Elements — Book I
Proposition 38

Formal Reconstruction — Technical Documentation

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Triangles on equal bases and between the same parallels

1 Introduction

Proposition I.38 continues the triangle part of Euclid's area block. The two triangles no longer share a base: ABC is on BC and DEF is on EF. The bases are congruent, the four base points are collinear, and the two top vertices lie on a line parallel to the common base carrier. Euclid concludes that the triangles are "equal." As in I.35–I.37, the relevant equality is synthetic equality of content, not congruence and not a numerical area value.

The formal statement therefore concludes `HilbertScissorsEquicomplementable`. It does not introduce an area function and it does not strengthen the conclusion to unconditional finite scissors equivalence. Hilbert's Chapter IV is the controlling source for this choice: his Theorem 46 states that triangles with equal bases and equal altitudes are equicomplementable, while he explicitly warns that the stronger equidecomposability assertion cannot be obtained in full generality without an Archimedean hypothesis.

The proof architecture, however, is not a literal formalization of Euclid's proof. The classical route constructs two surrounding parallelograms, invokes I.36, uses I.34 to identify the triangles as halves, and then cancels the common factor two. The Lean proof deliberately avoids this final cancellation. Instead it constructs a triangle GEF congruent to ABC, proves that G lies on the same upper parallel as A and D, applies I.37 to GEF and DEF, and finishes by transitivity of equicomplementability.

Thus I.38 is a particularly clean realization of Hartshorne's observation that, once the equal-content theory and I.37 are available, I.38 follows by transitivity. The work hidden by that sentence is geometric: one must build the correct congruent copy and prove that its third vertex has the required position.

2 The Classical Configuration

Let ABC and DEF be triangles on equal bases BC and EF. In the configuration used by the formal theorem the base points occur in the strict order

$$B - C - E - F,$$

encoded by

$$\text{Between}(B, C, E), \qquad \text{Between}(C, E, F).$$

The equality of bases is represented by segment congruence

$$BC \cong EF.$$

The top vertices A and D determine a line parallel to the base carrier:

$$AD \parallel BC.$$

These hypotheses are stronger than the informal phrase "on equal bases in the same parallels" because they expose the point order needed by the later synthetic argument. In particular, the Lean proof does not read the order of B, C, E, F from a drawing.

Figure 1 displays four stages of the proof. Panel (a) is the public configuration. Panel (b) shows the congruent-copy construction. Panel (c) displays the extension points and parallelogram that are hidden in the formal proof of the copy's position. Panel (d) is the nondegenerate branch in which I.37 is applied. The separate boundary case $G = D$ is handled directly in Lean and is not represented as a second picture.

3 Classical Proof

The classical proof follows the same pattern as I.37 but first uses I.36 to compare parallelograms on equal bases. In the standard configuration one constructs parallelograms around the two triangles so that a diagonal of each parallelogram cuts it into two congruent triangles. Wilkins describes the parallelograms as $GBCA$ and $DEFH$.

Because their bases BC and EF are equal and they lie in the same parallels, I.36 gives equality of the two parallelograms. Proposition I.34 then says that each diagonal divides its parallelogram into two congruent triangles. Thus each of the given triangles is one half of one of the equal parallelograms, and the classical conclusion is that the two halves are equal.

The dependency graph is therefore

$$\begin{gathered}
BC \cong EF, \quad \text{same parallels} \\
\Downarrow \quad \text{construct two parallelograms} \\
GBCA,\ DEFH \\
\Downarrow \quad \text{I.36} \\
GBCA = DEFH \\
\Downarrow \quad \text{I.34 twice} \\
\triangle ABC,\ \triangle DEF \text{ are corresponding halves} \\
\Downarrow \quad \text{“halves of equals are equal”} \\
\triangle ABC = \triangle DEF.
\end{gathered}$$

This is exactly the route visible in the Beeson proof corpus: the trace for `Prop38.prf` calls Proposition 36, Proposition 34 twice, converts triangle congruence to equality of figures, and ends with an explicit `halvesofequals` axiom. That trace is useful as a dependency audit, but it is not adopted as the architecture of the present proof.

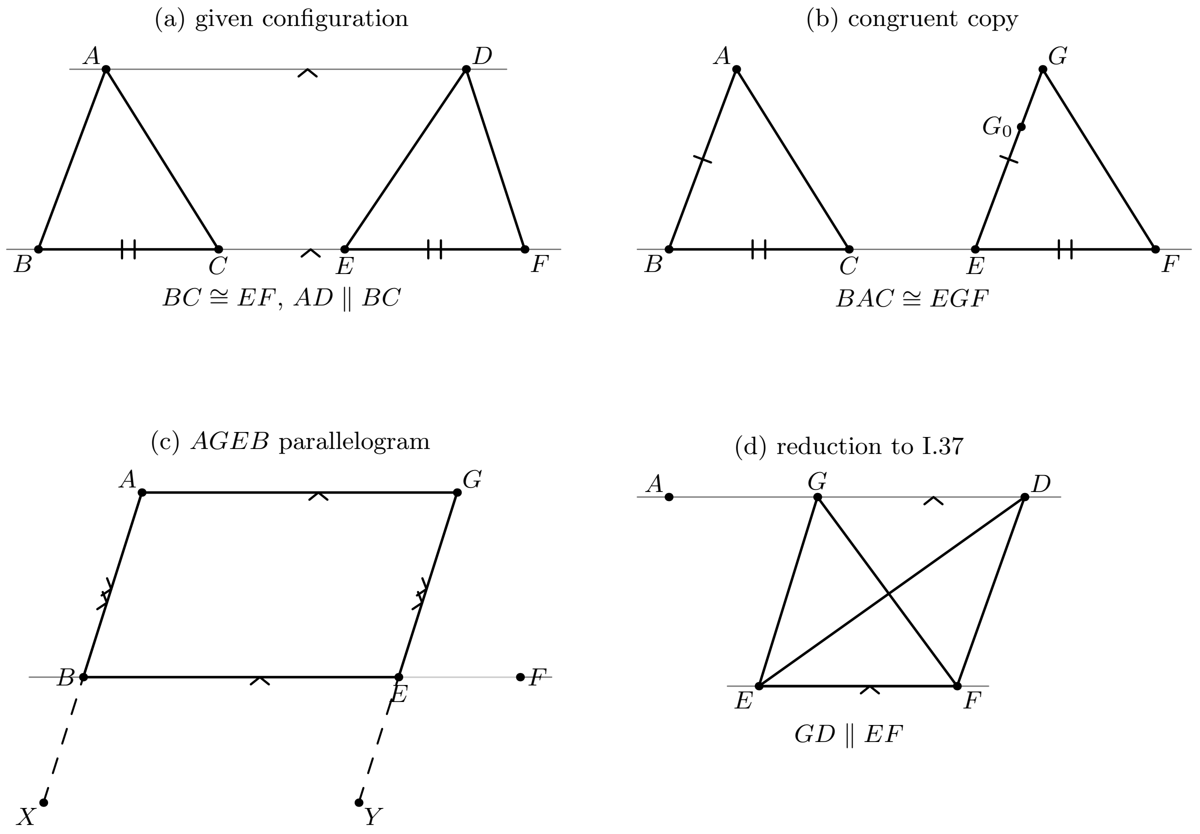

Figure 1: The principal geometric stages of the Lean reconstruction of I.38. (a) The given triangles have ordered collinear bases BC and EF, with $BC \cong EF$ and the top carrier parallel to the base carrier. (b) The angle at B is copied at E on the side containing A, and $EG \cong AB$ is laid off on the constructed ray; SAS gives $BAC \cong EGF$. The temporary point G_0 names the ray used for the layoff. (c) Extensions $A - B - X$ and $G - E - Y$ provide the oriented transversal configuration used with I.28; the resulting parallel and congruence data give the parallelogram $AGEB$. (d) The top carrier is identified with AD; in the branch $G \neq D$ one obtains $GD \parallel EF$, so I.37 compares GEF and DEF.

4 Equal Bases by Congruent Copy and Transitivity

Hartshorne's Section 22 gives the key conceptual simplification. After proving a version of I.37 that avoids the problematic halving principle, he states that I.38 follows by transitivity. This is possible because equal content is an equivalence relation and because equidecomposable figures have equal content.

In an informal affine picture, one can see the transitivity route immediately: translate a congruent copy of ABC onto the equal base EF. Its third vertex must lie on the same upper parallel, so I.37 compares that copied triangle with DEF. A Hilbert formalization cannot simply say "translate the triangle," because translation is not a primitive operation in the project. The copy has to be reconstructed from angle layoff, segment layoff, side-of-line data, SAS, and parallel uniqueness.

The Lean DAG is therefore

$$\begin{array}{c}
BC \cong EF,\ B - C - E - F,\ AD \parallel BC \\
\Downarrow \\
\text{construct } G \text{ with } BAC \cong EGF \\
\Downarrow \\
AGEB \text{ is a parallelogram} \\
\Downarrow \\
AG \parallel BC \text{ and } AG \text{ has the same carrier as } AD \\
\Downarrow \\
G = D \quad \text{or} \quad GD \parallel EF \\
\Downarrow \\
\begin{array}{l}
\triangle ABC \sim \triangle GEF \\
\triangle GEF \approx \triangle DEF \quad \text{by I.37} \\
\ \Downarrow \quad \text{transitivity} \\
\triangle ABC \approx \triangle DEF.
\end{array}
\end{array}$$

Here $\sim$ denotes the stronger relation `HilbertScissorsEq` produced by triangle congruence, while $\approx$ denotes `HilbertScissorsEquicomplementable`. The distinction is important: the first step is a genuine congruent replacement, while I.37 supplies only the weaker content relation that is correct in the non-Archimedean setting.

Hilbert's Theorem 46 supplies a broader control theorem: equal-base, equal-altitude triangles are equicomplementable. I.38 is exactly a Euclidean same-parallels instance of that geometry. The current library does not invoke Theorem 46 as a monolithic theorem; it realizes the equal-base case through the already completed I.37 and a congruent-copy reduction.

5 Proof Architecture of the Reconstruction

5.1 Step 1: recovering one common base carrier

The helper

```
i38_base_carrier
```

starts from $AD \parallel BC$ and uses `parallel_endpoints_sameSide` to obtain an actual Hilbert line `base` carrying B and C, together with SameSide information for A and D. The two strict betweenness hypotheses then place E and F on that same carrier by incidence transport.

Mathematically, the result packages the statement that the apparently separate segments BC and EF belong to one lower line, while both top vertices are off that line and lie on the same side of it. This is the first place where the formal proof exposes data that the classical phrase "the same parallels" suppresses.

The helper remains proposition-local. It is a useful normalization for this ordered six-point configuration, but it is not promoted to the general interface because its statement is tailored to $B - C - E - F$.

5.2 Step 2: nondegeneracy is derived, not assumed

The helper

`i38_nondegenerate`

proves

$$A, B, C \text{ noncollinear}, \qquad D, E, F \text{ noncollinear}.$$

Since B, C, E, F lie on the base carrier and SameSide data place A, D off that carrier, both conclusions are incidence consequences of the configuration.

This helper is logically small but important for the next stage. Hilbert's angle construction and SAS require explicit noncollinearity witnesses. The classical diagram treats these as evident; Lean requires them as proof terms. No new geometric hypothesis is added to the public theorem.

5.3 Step 3: constructing the congruent copy on the equal base

The central construction is

`i38_equal_base_copy`.

It first uses Hilbert's angle-construction axiom to copy the angle ABC at E, with the fixed ray directed through F and with the new ray chosen on the side of the base containing A. This creates a temporary point G_0 and an angle congruence of the form

$$\angle ABC \cong \angle FEG_0.$$

Next, Hilbert's segment construction lays off on the ray EG_0 a point G satisfying

$$EG \cong AB.$$

The relation `HilbertSameRay` between G_0 and G is then used twice: first to preserve the side-of-line choice, and second to replace G_0 by G in the copied angle. Thus the construction retains the orientation selected by the angle axiom instead of merely obtaining an unoriented congruent angle.

With the public hypothesis $BC \cong EF$, SAS is applied to the triangles in the formal orientation

$$BAC \quad \text{and} \quad EGF.$$

The result is

$$\text{TriangleCongruenceResult}(BAC, EGF).$$

Panel (b) of Figure 1 records this stage. The point G_0 is not part of the final geometry; it exists solely to name the constructed ray on which the final segment is laid off.

This helper is proposition-local, but it reveals a reusable proof pattern: angle layoff + segment layoff + SameRay transport + SAS gives a controlled oriented copy of a triangle on a prescribed congruent base.

5.4 Step 4: proving that the copy is a parallelogram

The helper

`i38_copy_parallelogram`

contains the most substantial hidden positional argument of I.38. The desired conclusion is

$$AGEB \text{ is a parallelogram.}$$

At an affine level this is the statement that the congruent copy on EF is a translation of the original triangle. The project does not use translation as a primitive notion, so the claim is rebuilt synthetically.

First GE is extended through E to a point Y, giving $G - E - Y$, and AB is extended through B to a point X, giving $A - B - X$. From the common base carrier and the SameSide choice of G, the proof derives the required SameSide/OppositeSide relations for the transversal configuration. The lower order $B - C - E - F$ also yields the derived order $F - E - B$.

The copied angle is normalized from the ray BC to the ray BE using `HilbertSameRay`; then the unordered angle representation is reoriented to exactly the corresponding-angle shape expected by I.28. Proposition I.28 is applied in its neutral corresponding-angle form to obtain parallelism of the extended side lines. Collinearity transport then gives

$$AB \parallel GE.$$

SAS had already supplied the corresponding side congruence

$$AB \cong GE.$$

Together with the fact that A and G lie on the same side of the lower carrier, these data satisfy `OnePairParallelCongruent`. The reusable criterion `OnePairParallelCongruentCriterion` therefore yields

$$AGEB \text{ parallelogram.}$$

Panel (c) of Figure 1 shows the extension witnesses X, Y that make this argument formally oriented.

This helper is local to I.38. The underlying one-pair parallelogram criterion is reusable Interface infrastructure and corresponds closely to the stronger parallelogram theorem isolated in the Borsuk–Szmielew development.

5.5 Step 5: identifying the upper carrier

The helper

```
i38_copy_upper_position
```

turns the parallelogram conclusion into the exact hypothesis needed by I.37. From $AGEB$ it obtains

$$AG \parallel EB.$$

Because E, B, C are collinear, parallel transport gives

$$AG \parallel BC.$$

But the public hypothesis already says

$$AD \parallel BC.$$

The crucial point is now uniqueness of the Euclidean parallel. If the extensional point-lines AG and AD were distinct, the developed transitivity result `hilbert_parallel_transitive_distinct` would make them parallel to one another. This is impossible because they both contain A. Hence the two point-line carriers are equal, and G lies on the upper carrier through A, D.

The proof then separates a boundary case that a drawing normally ignores:

$$G = D$$

or

$$G \neq D.$$

If $G = D$, the copied triangle already has the desired top vertex. Otherwise, parallelism is transported along the common upper and lower carriers to obtain

$$GD \parallel EF.$$

Panel (d) of Figure 1 depicts this nondegenerate branch.

This step is the clearest point at which the Euclidean uniqueness theory of parallels becomes indispensable. The earlier angle construction, segment construction, side-of-line transport, SAS, and I.28 criterion are neutral.

5.6 Step 6: the content proof is now short

The final theorem converts the triangle congruence certificate into `HilbertScissorsEq` using `scissors_congruent`. Because triangle terms are represented as unordered multisets of vertices, two applications of `scissors_triangle_swap12` normalize

$$BAC \sim EGF$$

to

$$ABC \sim GEF.$$

This is a representation step only; it changes no geometry.

The stronger scissors equality is embedded into equicomplementability by the existing helper

`equicomplementable_of_scissorsEq`.

The two position cases then close as follows.

If $G = D$, the copied triangle is already DEF, so the congruent replacement finishes the proof. If $G \neq D$, the parallelism $GD \parallel EF$ is exactly the hypothesis of Proposition I.37 for the two triangles GEF and DEF. Thus

$$GEF \approx DEF.$$

Finally `equicomplementable_trans` composes

$$ABC \approx GEF, \qquad GEF \approx DEF$$

to obtain the public conclusion.

This is the content stage of the proof. All of the preceding line, ray, SameSide, extension, and parallel arguments exist solely to put the copied triangle into the configuration in which I.37 can be called.

6 Lean Formalization

The current Proposition file imports

```
import CGJteamLab.Proposition28
import CGJteamLab.Proposition37
```

Thus I.28 is the direct neutral parallel criterion used inside the geometric copy argument, while I.37 is the direct content theorem used at the end.

The public theorem is

```
theorem euclid_proposition_38
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C D E F : Geo.Point)
    (hAD_BC : Geo.Parallel A D B C)
    (hBCE : Geo.Between B C E)
    (hCEF : Geo.Between C E F)
    (hBC_EF : Geo.Congruent B C E F) :
    HilbertScissorsEquicomplementable Geo
      (hilbertScissorsTriangle Geo A B C)
      (hilbertScissorsTriangle Geo D E F) := by
  ...
```

The ellipsis here is documentation notation only; the repository theorem is fully proved and contains no `sorry`.

The proposition-local proof path is represented by

```
i38_base_carrier
i38_nondegenerate
i38_equal_base_copy
i38_copy_parallelogram
i38_copy_upper_position
euclid_proposition_38
```

The relatively short list should not be mistaken for a trivial proof. The middle three helpers package the substantial geometry: orientation of the copied triangle, the I.28 extension configuration, construction of the parallelogram, and identification of the upper carrier.

At the top level the final theorem is intentionally small. It consumes the geometric package returned by `i38_copy_upper_position`, converts the congruent copy to a scissors equivalence, and dispatches the two position cases to either reflexive transport or I.37 plus transitivity.

7 Relationship with Earlier Propositions

7.1 Proposition I.28

I.28 is a direct formal dependency. The helper `i38_copy_parallelogram` uses the theorem

`euclid_proposition_28_corresponding`

after explicitly constructing the extension points and the opposite-side configuration required by the transversal theorem. This is a neutral use: equal corresponding angles imply parallelism without invoking the Euclidean uniqueness axiom.

7.2 Proposition I.33

The final I.38 file does not call `euclid_proposition_33` by name, but its mathematical role is present in reusable form. The Interface theorem `OnePairParallelCongruentCriterion` says that one oriented pair of parallel and congruent opposite sides determines a parallelogram. This is the structural content needed to turn $AB \parallel GE$ and $AB \cong GE$ into the parallelogram $AGEB$.

7.3 Proposition I.34

I.34 belongs to Euclid's classical proof: it supplies the diagonal congruence showing that each surrounding parallelogram is double its triangle. The Lean proof does not call I.34. Avoiding that step is precisely what removes the need for a general "halves of equals" principle.

7.4 Proposition I.36

I.36 is the decisive earlier proposition in the classical DAG. It compares the two constructed parallelograms on the equal bases BC and EF. Nevertheless, it is not an actual dependency of the final Lean proof. The formal route replaces the I.36-plus-halving argument by a congruent-copy reduction to I.37.

This difference is structurally important. It confirms that the formal Book I dependency graph is not forced to reproduce Euclid's proposition order even when the public statement remains faithful to Euclid.

7.5 Proposition I.37

I.37 is the decisive content dependency in the actual Lean proof. Once the copy point G has been placed on the upper carrier and the nondegenerate branch gives $GD \parallel EF$, I.37 applies immediately to the triangles GEF and DEF on the common base EF.

Thus the formal relationship is

$$\text{congruent replacement} \quad + \quad \text{I.37} \quad + \quad \text{transitivity.}$$

This is the exact formal meaning of Hartshorne's compressed statement that I.38 follows by transitivity.

8 Hilbert and Book Zero Components Used

8.1 Incidence, order, and side-of-line infrastructure

The proof uses the developed Hilbert language for

- strict betweenness and its incidence consequences;

- propagation of collinearity to a fixed carrier line;
- noncollinearity from a point lying off a line;
- line existence and uniqueness through two distinct points;
- SameRay transport;
- SameSide and OppositeSide symmetry and transitivity;
- extension of a segment beyond an endpoint.

These mechanisms are responsible for most of the positional detail absent from the classical diagram.

8.2 Congruence construction and SAS

Hilbert's congruence layer supplies two genuine constructions: `angle_construction` creates the oriented ray at E, and `segment_construction` lays off $EG \cong AB$ on that ray. The theorem `hilbert_sameRay_points_sameSide` ensures that replacing G_0 by the final layoff point G preserves the chosen half-plane.

SAS then gives the full triangle-congruence package

$$BAC \cong EGF.$$

No circle construction or continuity axiom is used in this stage.

8.3 Parallel and parallelogram infrastructure

The geometric reduction consumes

- I.28 corresponding-angle parallelism;
- parallel transport along collinear carriers;
- `OnePairParallelCongruentCriterion`;
- Euclidean transitivity and uniqueness of parallels;
- the extensional `PointLine` representation used to identify the upper carrier.

The distinction between neutral existence/angle criteria and Euclidean uniqueness is visible here rather than hidden in a generic "parallel" tactic.

8.4 Scissors and content infrastructure

No new scissors constructor is needed for I.38. The proof uses only established operations:

- `scissors_congruent`;
- triangle-term permutation lemmas;

- equicomplementable_of_scissorsEq;
- equicomplementable_trans;
- Proposition I.37.

This is a strong confirmation that the term-level content interface introduced at I.35 is adequate for I.38.

9 Formalization Notes

9.1 Geometry, representation, and content

I.38 again separates cleanly into three layers.

Geometry. The proof constructs the copied ray and segment, proves the copied triangle congruent, derives the parallelogram $AGEB$, identifies the common upper carrier, and transports the lower carrier from BC to EF.

Representation. Angles have to be reversed or transported along SameRay relations to match the argument order required by I.28 and SAS. At the end, the congruent triangles appear as BAC and EGF, while the public scissors terms are ABC and GEF. The triangle swap lemmas normalize this representation without adding geometric content.

Content. Only after the geometry is complete does the proof enter the scissors layer. Congruence gives a strong `HilbertScissorsEq`; I.37 gives `HilbertScissorsEquicomplementable`; transitivity composes them. No subtraction, cancellation, numerical area, or new polygon semantics is needed.

9.2 Hidden configuration data

The formal proof explicitly establishes the following facts instead of reading them from Figure 1:

- B, C, E, F lie on one Hilbert line;
- A, D lie off that line and on the same side of it;
- both original triangles are noncollinear;
- the copied point G_0 is on the chosen side of the base;
- G is on the same ray from E as G_0;
- moving from G_0 to G preserves SameSide data and the copied angle;
- $G - E - Y$ and $A - B - X$ are strict extensions;
- the extension point Y is on the opposite side of the base from G and therefore from A;
- the order $B - C - E - F$ implies the transversal order $F - E - B$;

- the exact angle orientation required by I.28;
- equality of the extensional carriers AG and AD;
- the boundary case $G = D$ separately from the strict case $G \neq D$.

The last two items are particularly significant: a standard affine drawing makes G look automatically located on the upper line, but in Hilbert geometry that position is a theorem obtained from Euclidean parallel uniqueness.

9.3 Axiomatic status

The public theorem assumes

```
[HilbertEuclideanPlane Geo]
```

and is therefore Euclidean in the current project API.

The proof nevertheless contains a substantial neutral prefix. Incidence, strict order, SameSide/OppositeSide reasoning, angle construction, segment construction, SameRay transport, SAS, and the corresponding-angle direction of I.28 do not require the Euclidean uniqueness of parallels.

Group IV enters when the copied side data are converted into the full parallelogram structure and, most visibly, when the proof identifies the two upper carriers through Euclidean transitivity/uniqueness of parallels. The final call to I.37 also inherits the Euclidean-plane instance.

No new parallel axiom is introduced in I.38. No continuity or Archimedean axiom is added. The construction of G is an existence argument inside the neutral congruence/order layer; the specifically Euclidean force lies in the uniqueness and structural consequences of parallelism.

9.4 Why the conclusion remains equicomplementability

Hilbert's Theorem 46 is the decisive semantic control. Equal-base, equal-altitude triangles are equicomplementable. Hilbert explicitly notes that without the Archimedean axiom one cannot in general strengthen this to finite equidecomposability.

The Lean proof reflects that distinction exactly. The replacement $ABC \rightsquigarrow GEF$ is a direct scissors equality because the two triangles are congruent. The subsequent comparison $GEF \rightsquigarrow DEF$ comes from I.37 and is only equicomplementability. Therefore the final theorem correctly returns the weaker relation.

9.5 Reusable helpers and proposition-local helpers

I.38 introduces no new reusable theorem in `HilbertInterface`, Book Zero, or `HilbertScissors`. Its new declarations are all prefixed `i38_` and remain proposition-local.

Their mathematical roles are nevertheless worth recording:

- `i38_base_carrier`: ordered lower-line normalization;

- `i38_nondegenerate`: noncollinearity consequences;
- `i38_equal_base_copy`: oriented congruent-copy construction;
- `i38_copy_parallelogram`: corresponding-angle and one-pair parallelogram argument;
- `i38_copy_upper_position`: Euclidean carrier identification and reduction to the I.37 configuration.

If the congruent-copy pattern recurs in later propositions, the third and fourth helpers are candidates for refactoring. At I.38 itself there is not yet enough evidence to enlarge the general Interface merely to shorten one proposition.

9.6 Source audit

The source audit assigns distinct roles to the standard references.

Euclid's classical proof, as recorded by Wilkins, uses two constructed parallelograms, I.36, I.34 twice, and the concluding halving principle. The Beeson proof corpus confirms the same dependency at a formal trace level and makes the final `halvesofequals` assumption explicit.

Hilbert, Chapter IV, Section 19, supplies the semantic control. Theorems 44–46 distinguish equidecomposability from equicomplementability and culminate in the theorem that equal-base, equal-altitude triangles are equicomplementable.

Hartshorne, Section 22, is the most direct architectural source for the chosen proof. After replacing Euclid's I.37 halving argument by a proof valid for equal content without property (Z), he states that I.38 follows by transitivity. The present Lean proof makes the geometric content of that transitivity step explicit by constructing GEF.

Borsuk–Szmielew are not the source for the equal-content semantics at this stage, but their Euclidean parallelogram theory is directly relevant to the one-pair parallel-plus-congruence criterion consumed by the proof. Their organization supports keeping this criterion in reusable interface infrastructure.

Greenberg remains a control source for the semantic boundary. He distinguishes numerical area from equal content and points to Hartshorne for the full synthetic theory rather than developing it himself. This supports the decision not to replace the scissors relation by a numerical area function.

The Euclid/Joyce edition and the project Book I source compilation remain statement and historical cross-checks; neither is used to force the Lean proof into the classical proposition DAG.

10 Dependency Summary

The actual formal dependency chain is

Hilbert incidence, order, congruence, and Euclidean parallel axioms

$\Downarrow$

HilbertInterface / Book Zero line, ray, side-of-line, congruence theory

$\Downarrow$

angle construction + segment construction + SAS

$\Downarrow$

`i38_equal_base_copy`

$\Downarrow$

I.28 corresponding-angle criterion + parallelogram infrastructure

$\Downarrow$

`i38_copy_parallelogram`

$\Downarrow$

Euclidean parallel uniqueness / carrier identification

$\Downarrow$

`i38_copy_upper_position`

$\Downarrow$

congruent scissors replacement + Proposition I.37

$\Downarrow$

`equicomplementable_trans`

$\Downarrow$

`euclid_proposition_38`.

The historical and formal DAGs are therefore sharply different:

$$\begin{aligned} &\text{Euclid:} && \text{construct parallelograms} \to I.36 \to I.34 \text{ twice} \to \text{halving,} \\ &\text{Lean:} && \text{congruent copy} \to \text{upper-carrier proof} \to I.37 \to \text{transitivity.} \end{aligned}$$

New geometric axiom:	no
New proposition-local helpers:	yes
New reusable Interface/Scissors helper:	no
New Book Zero theorem:	no
New numerical area or polygon semantics:	no
Public theorem compiles:	yes
Full `lake build`:	clean
`sorry` in Proposition38.lean:	no

11 Conclusion

Proposition I.38 is classically presented as the equal-base analogue of I.37, but its formal reconstruction exposes a more useful structure. The content part of the theorem is almost entirely inherited from I.37 and transitivity. The new work is geometric: construct an oriented congruent copy on the second base, prove that the copy is related to the original triangle by a parallelogram, and use Euclidean parallel uniqueness to place its top vertex on the required upper carrier.

This explains why a short classical proposition can generate a substantial Lean development. The extra code is not a hidden area calculation. It is the formal cost of orientation, side-of-line data, ray replacement, extension points, carrier transport, and uniqueness of parallels – all information that a conventional diagram suppresses.

At the same time, I.38 is evidence that the content architecture introduced at I.35 has stabilized. No new scissors operation, polygon semantics, numerical area, or cancellation principle is required. Congruence supplies the strong local replacement $ABC \sim GEF$; I.37 supplies the equal-content comparison $GEF \approx DEF$; and transitivity closes the theorem. In that precise sense, the final proof realizes Hartshorne's concise diagnosis: I.38 follows by transitivity, once the geometry needed to make the intermediate triangle explicit has been proved.